

Jelly Jump — Teaching Guide (Pseudocode + Build Order)

Game link: <https://play.codecampworld.com/play.html?game=928RIXF>

How to play: Press/hold on the level to charge a jump → release to jump the slime → hit springs to bounce left → reach the flag in as few attempts as possible.

This is the order the lesson introduces the code in. Each step below is a checkpoint you can pause on to check students are with you.

1. Jelly Actor — Setup (already in the starter code)

ON CREATE:

```
    camera follows this actor
    lock rotation (don't flip around)
    scene gravity = 800 (fall speed)
    charging = false          // are we currently charging a
jump?
    speed = default          // how hard we jump
```

2. Jelly Actor — Charge (students build this)

Goal: when the slime is on the ground AND the mouse is held down, start charging.

ON UPDATE:

```
    IF this.isBlockedDown() AND mouse is down:
        charge()
```

Talking point: nothing happens yet because `charging` starts `false` and nothing sets it `true`. That's the next step.

3. Jelly Actor — Reset (already provided)

```
FUNCTION reset():
    charging = true
    speed = default
```

Wire `reset()` into the charge check so charging actually turns on:

ON UPDATE:

```
    IF this.isBlockedDown() AND mouse is down:
        IF NOT charging:
            reset()
        ELSE IF this.getYScale() > 0.2:    // still squashing
down = still charging
            charge()
```

4. Jelly Actor — Jump / Release (students build this)

```
FUNCTION release():  
    IF charging:  
        jump upward  
        set x speed (move right)  
        reset y-scale back to 1 (undo the squash animation)  
        charging = false
```

Add the "let go" branch to the update loop:

```
ON UPDATE:  
    IF (this.isBlockedDown()) AND mouse is down:  
        IF NOT charging: reset()  
        ELSE IF getYScale() > 0.2: charge()  
    ELSE IF this.isBlockedDown():  
        release()
```

Common bug #1 — can only jump once: `isBlockedDown()` alone stops registering once the slime has moved. Fix by also checking for a side collision:

```
IF (this.isBlockedDown() OR this.didCollideOn("bottom")) AND  
mouse is down:  
    ...
```

5. Jelly Actor — Restart (students build this)

Two versions appear in the lesson — start simple, then upgrade:

```
# v1 — simple restart  
ON UPDATE (always):  
    IF x-position > scene width OR y-position > scene height:  
        restart the current scene  
# v2 — keeps score across restarts (final version)  
ON UPDATE (always):  
    IF x-position > scene width OR y-position > scene height:  
        x speed = 0  
        x position = startX  
        y position = startY  
        attempts = attempts + 1
```

6. Flag Actor — Setup (already provided)

```
ON CREATE:  
    no gravity, can't be knocked around by other actors
```

7. Flag Actor — Win (students build this)

```
ON HIT by "Jelly":  
    play "firework1" animation
```

```
after 1 second: go to next scene
set x speed = 0 for the actor that hit it
```

8. Spring Actor (students build this, then place it in Level 3)

ON CREATE:

```
no gravity
sensor = true           // doesn't physically block, just
detects touch
direction = 270         // faces left
```

On the Jelly actor:

ON HIT "Spring":

```
set x speed = -300 for this // bounce left
```

Placement note: drag a Spring into Level 3 and position it on the wall the student is meant to bounce off of.

9. Ice / Sliding (students build this)

NEW GAME VARIABLE: iceLevel

Level 1 scene setup:

ON SCENE CREATED (once):

```
iceLevel = false
```

Jelly actor:

ON HIT a tile:

```
IF NOT iceLevel:
```

```
    x speed = 0 // stop sliding on grass
```

Talking point: levels 2 and 3 are ice levels, so `iceLevel` should be `true` there — sliding is only meant to happen on ice.

10. Attempt Counter (students build this)

Level 1 setup, 3 new variables:

```
attempts = 1
```

```
startX = slime's starting x position
```

```
startY = slime's starting y position
```

These feed the `v2 restart` logic in step 5 — every fall increments `attempts` instead of fully restarting the scene.

Suggested teaching flow (checkpoints)

1. Demo the finished game, discuss objective and controls.
2. Walk through existing Setup/Reset code together (steps 1, 3, 6).

3. Build Charge → wire in Reset → test (steps 2–3). **Checkpoint: charging works but jump does nothing yet.**
4. Build Release, test jump. **Checkpoint: can jump once.**
5. Diagnose "can only jump once" as a class, apply the OR-collision fix (step 4 bug).
6. Build Restart v1, test falling off-screen (step 5).
7. Build Flag Win condition, test finishing a level (step 7).
8. Build Spring, place in Level 3, test bounce (step 8).
9. Add iceLevel + sliding fix (step 9).
10. Add attempts counter and upgrade Restart to v2 (steps 5 & 10).